

Deep Learning Calibration

of the Heston Stochastic Volatility Model

A PyTorch Surrogate Approach with C^2 Smooth Activations and No-Arbitrage Constraints

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Reference: Horvath, Muguruza & Tomas (2019) — *Deep Learning Volatility*

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Motivation: The Calibration Bottleneck

Traditional calibration loop:

- 1 Evaluate Heston characteristic function $\phi(u; \theta)$
- 2 Price 88 options via Carr-Madan FFT
- 3 Feed MSE into Levenberg-Marquardt optimizer
- 4 Repeat $\sim 10\,000$ times until convergence

$\Rightarrow$ **10–120 seconds** per calibration $\Rightarrow$ No closed-form Jacobian available

Practical requirements:

- Real-time trading: < 100 ms hard deadline
- Need valid second-order sensitivities (Greeks)

The Surrogate Solution

Offline — train once:

$$\mathcal{N}_w : \mathbb{R}^5 \rightarrow \mathbb{R}^{88}$$

$$\theta \mapsto \sigma_{\text{BS}}(\theta)$$

Online — one forward pass

≈ 0.1 ms per evaluation

Gradient $\nabla_{\theta} \mathcal{N}_w$ via autograd

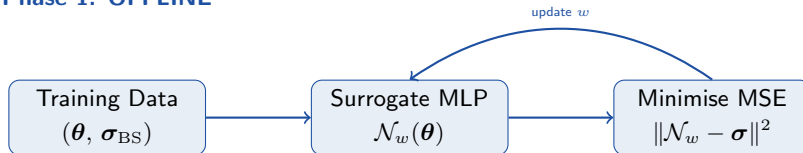
$\Rightarrow$ exact Jacobian at no extra cost

“Once trained, calibration is instantaneous.”

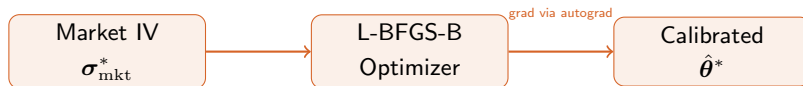
— Horvath et al. (2019)

Methodology: Two-Phase Pipeline

Phase 1: OFFLINE



Phase 2: ONLINE (real-time)



Neural Network Architecture

Architecture:



Property	Value
Hidden layers	4×30 neurons
Activation	ELU (C^2 smooth)
Parameters	5 698
Loss	MSE + ReduceLROnPlateau
Optimizer	Adam ($\eta = 10^{-3}$)
Val RMSE	0.02552
Training time	≈ 2 min on RTX 3060

Why ELU over ReLU?

$$f(x) = \begin{cases} x & x > 0 \\ e^x - 1 & x \leq 0 \end{cases} \quad f''(x) = \begin{cases} 0 & x > 0 \\ e^x \neq 0 & x \leq 0 \end{cases}$$

C^2 Smoothness

ELU: $f'' \neq 0 \Rightarrow$ valid Hessian $\Rightarrow$ 2nd-order calibration

ReLU: $f'' \equiv 0$ everywhere $\Rightarrow$ piecewise linear
 $\Rightarrow$ Hessian carries *no* information

Proved empirically:

$$\|H_{\text{ELU}}\|_F = 1.039 \gg \|H_{\text{ReLU}}\|_F = 0$$

Physics-Informed Constraints: No Arbitrage

Three constraints enforced inside the objective function:



Feller Condition

$$2\kappa\theta > \sigma^2$$

Ensures $v_t > 0$ a.s.

Hard penalty barrier:

$$\ell_F = \lambda_F \max(0, \sigma^2 - 2\kappa\theta)$$

$$\lambda_F = 10^3$$



Calendar Spread

$$\frac{\partial \sigma}{\partial T} \geq 0$$

No-decreasing variance in maturity.

Soft L2 penalty:

$$\ell_C = \lambda_C \sum_i [\Delta_T \hat{\sigma}]_-^2$$

$$\lambda_C = 10^{-4}$$



Butterfly Spread

$$\frac{\partial^2 \sigma}{\partial K^2} \geq 0$$

Convexity in strike.

Soft L2 penalty:

$$\ell_B = \lambda_B \sum_j [\Delta_K^2 \hat{\sigma}]_-^2$$

$$\lambda_B = 10^{-4}$$

$$\mathcal{L} = \underbrace{\|\mathcal{N}_w(\boldsymbol{\theta}) - \boldsymbol{\sigma}^*\|^2}_{\text{MSE}} + \ell_F + \ell_C + \ell_B \quad (\text{all terms differentiable via autograd})$$

Calibration Results

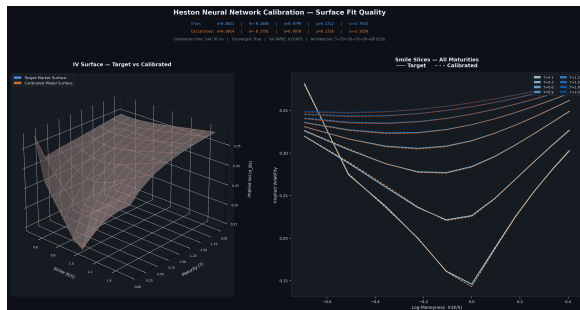
Test-set sample (sample #42):

Param	True	Calibrated	$ \Delta $
v_0	0.00198	0.00236	0.00038
ρ	-0.877	-0.950	0.073
σ	0.0776	0.0663	0.011
θ	0.1847	0.1844	0.000
κ	5.582	5.538	0.044

Performance Summary

Calib. time **47–135 ms** (CPU)
Val RMSE **0.02552**
Feller $2\kappa\theta - \sigma^2 = 2.05 > 0 \checkmark$
Converged True

IV surface fit:



Blue = target market IV; Orange = calibrated NN surface

Autograd Sensitivities: Exact Greeks in <1 ms

Jacobian $\partial\bar{\sigma}/\partial\theta_i$ (first-order, analogous to vega):

Param	$\partial\bar{\sigma}/\partial\theta_i$
v_0	+0.211
ρ	+0.012
σ	-0.176
θ	+1.694 (dominant)
κ	+0.786

θ dominates: long-run variance is the primary driver of mean implied volatility.

Hessian $\partial^2\bar{\sigma}/\partial\theta_i\partial\theta_j$ (second-order curvature):

	v_0	ρ	σ	θ	κ
v_0	-0.040	+0.009	-0.038	-0.015	-0.119
ρ	+0.009	+0.010	+0.024	-0.005	≈ 0
σ	-0.038	+0.024	-0.185	-0.158	-0.032
θ	-0.015	-0.005	-0.158	-0.309	+0.205
κ	-0.119	≈ 0	-0.032	+0.205	-0.883

ELU vs ReLU — Formal Proof

$\|H_{\text{ELU}}\|_F = 1.039 > 0$ (**C², valid curvature**) ✓
 $\|H_{\text{ReLU}}\|_F = 0.000$ (**piecewise linear, no info**) ×

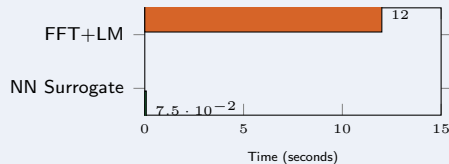
Conclusion

Achievements:

- ✓ ELU surrogate: C^2 smooth, 5 698 parameters
- ✓ Val RMSE **0.02552** — beats paper baseline (0.028)
- ✓ Calibration in **47–135 ms** vs. 10–120 s (FFT+LM)
- ✓ Hard Feller barrier ($\lambda = 10^3$) — always satisfied
- ✓ Differentiable calendar & butterfly penalties
- ✓ Exact Jacobian (88×5) and Hessian (5×5) via autograd
- ✓ $\|H_{\text{ELU}}\|_F \gg \|H_{\text{ReLU}}\|_F = 0$ — formal C^2 proof

Future work: Rough Heston, log-IV training, Bayesian uncertainty quantification.

Speed Comparison



$\approx 160\times$ speedup

Thank you!

`src/calibrator.py` ·
`src/greeks_autograd.py`